INCH-POUND

A-A-52147
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COMMERCIAL ITEM DESCRIPTION

SANDWICH UNIT, MECHANICALLY REFRIGERATED

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-S-43852.

1. ABSTRACT

1.1 Abstract. This commercial item description covers mechanically refrigerated sandwich units in the following types, styles, and sizes (see 5.2):

Type I - water cooled condenser
Type II - air cooled condenser

Style A - with four swivel casters
Style B - with adjustable legs

Size 1 - 6 cubic foot
Size 2 - 9 cubic foot
Size 3 - 11 cubic foot

1.2 CID based part identification number (PIN). A document based PIN to identify types, styles, and sizes of items is included in section 5. This identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor except as information on "Government unique" data nameplates (see 5.4).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-WEE, Natick, MA 01760-5018.

AMSC N/A FSC 7310

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2. SALIENT CHARACTERISTICS

2.1 Design and construction. The sandwich unit shall be a completely self-contained, one or multiple door, mechanically refrigerated, unit; air cooled or water cooled with a two-shelf (minimum) food storage compartment and a hinged or snap-on removable covered food pan top(s). A full length, removable, synthetic cutting board 8 inches (minimum) width shall be mounted to the top of the unit. The refrigeration system shall automatically maintain the interior of the sandwich unit at 38°F ± 2°F, in an ambient temperature of 100°F ± 2°F. The maximum overall dimensions shall be in accordance with Table I for the sizes listed therein.

TABLE I. Overall dimensions (inches)

Size	Max. length	Max. depth	Working height (adjustable)	Food pans surface area (square inches)
1	38	34	35-38	360
2	50	34	35-38	480
3	62	34	35-38	720

The refrigerated storage area shall be a minimum of 6 cubic feet for the size 1, 8 cubic feet for the size 2, and 11 cubic feet for the size 3. The interior and exterior of the cabinets and the door(s) shall be fabricated of 22 gauge (minimum) 300 series stainless steel conforming to ASTM A167. The space between the exterior sheet metal and the interior sheet metal of the cabinet and the door(s) shall be completely filled with insulation. Insulation shall be adequate to prevent formation of condensation on the exterior of the cabinet when ambient air conditions are 95°F ± 2°F and 94% relative humidity and the interior of the refrigerated section is at 38°F ± 2°F.

2.2 Cabinet top. The top of the cabinet for each sandwich unit shall accommodate the number and size of food pans specified in table II. The food pans and adaptor bars specified in table II shall be provided with the sandwich units. The opening in the top of the cabinet for the food pans specified shall be completely covered when all pans are inserted in normal use position.

TABLE II. Pan sets and adaptor bars

Pan size	Pan capacity (quarts)	Pan depth (inches)	Quantity per sandwich unit		
			Size 1	Size 2	Size 3
One-sixth	1-1/4	2-1/2	8 (min)	-	-
One-quarter	2	2-1/2	-	4	4
One-third	2-3/4	2-1/2	-	3	6
Adaptor bar(s)	-	-	1 (min)	1	2

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The area over the pans shall have a full accessed hinged or snap-on cover. This cover shall be easily removed for cleaning purposes. Hinges for the cover shall be of a type to prevent the cover from becoming detached when opening or closing the cover. The top shall also be provided with an 8 inch full length removable synthetic cutting board, and shall have a smooth top surface free from openings. The top of the cabinet for the sizes 2 and 3 sandwich units shall contain a toaster or appliance section with a back splash and a cutout for an appliance receptacle. A stainless steel cover shall be provided and installed over the cutout for the appliance receptacle.

2.3 Tray rails. When specified, a stainless steel drop-type tray rail, with supporting hardware, shall be supplied with each sandwich unit on the partron's side (see 5.2). The tray rail shall extend the full length of the sandwich unit and the height of the tray rail from the floor shall be adjustable from 34 inches to 35 inches.

2.4 Casters, legs, and bumpers. Style A sandwich units shall be furnished with four full swivel type casters that shall have hard rubber or canvas-phenolic resin impregnated wheels. A minimum of two casters shall be equipped with a foot operated brake device which shall hold the cabinet stationary. Style B sandwich units shall be furnished with NSF International approved 6 inch adjustable legs. Style A sandwich units shall be equipped with bumpers to cushion the impact of collisions and prevent damage to equipment. Bumpers shall be capable of being removed and replaced when necessary.

2.5 Refrigeration unit. The refrigeration unit shall be an integral part of the complete sandwich unit. The refrigeration unit shall be completely self-contained, precharged, and wired with a hermetically sealed compressor unit that shall be capable of maintaining a cabinet interior temperature of $38^{\circ}F \pm 2^{\circ}F$ in an ambient of $100^{\circ}F \pm 2^{\circ}F$. Unless otherwise specified, the refrigeration unit shall operate on a 120 V, 60 Hz, single-phase electrical system (see 5.2), and shall be provided with a six foot, three-wire cord conforming to Underwriters Laboratories (UL) Standard No. 62 and with a three-prong grounding type plug conforming to the National Electrical Manufacturers Association (NEMA) Standard No. WD-1. When 50 Hz frequency is specified, the motor and wiring shall be specifically designed for operation on that frequency.

2.5.1 Ozone depleting substances. Class I ozone depleting substances, in accordance with section 602(a) of the Clean Air Act Amendments 1990, 42 USC 7671a(a) as implemented by 40 CFR 82, Protection of Stratospheric Ozone, shall not be used for refrigerant gases, or for any other purpose, or employed in any process associated with the manufacture of any component parts for this item. See the 5.6 for additional information.

2.6 Standard compliance. The sandwich unit shall comply with UL Standard No. 62 and No. 471, NEMA WD-1, and NSF International Standard No. 2.

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2.7 Data nameplate ("Government unique" requirement). The sandwich unit shall be furnished with a data nameplate in accordance with the applicable requirements of MIL-STD-130 except the requirements for: (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply. The data nameplate shall be made of minimum 20 gauge corrosion-resisting metal and attached to the sandwich unit by rivets, screws, or welding in such a manner as to meet the applicable NSF International sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

Contract Number
National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List Only if Different from Manufacturer)
Manufacturer's Model and Serial Number
Government Approved Manual Number

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the sandwich unit.

2.8 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

2.9 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this commercial item description unless otherwise specified.

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2.10 Workmanship. The finished sandwich unit shall be complete, clean, and free of fabrication marks, scratches, dents, sharp edges, sharp corners, burrs, slivers, or visible evidence of rust. Rivets, when used, shall be tight and the rivet heads and peened ends shall be round and concentric and in full contact with the joined metal surfaces. There shall be no empty rivet holes. Welds shall be clean, continuous, sound, and free of flux, pits, buildups, undercuts, and shall be ground so as to blend with the adjacent surfaces. Spot welds shall be free from pits, burn throughs, and flashes.

3. QUALITY ASSURANCE

3.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics and requirements of this commercial item description and except for those Government unique requirements specified, conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace or that had successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Compliance. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the sandwich unit conforms to the requirements of UL Standard No. 62, No. 471, NEMA WD-1, and NSF International Standard No. 2, as applicable.

3.2.1 Underwriters Laboratories, Inc. (UL). Acceptable evidence of meeting the requirements of UL Standard No. 62 and No. 471 shall be the UL listing mark, or a certified test report from a recognized independent laboratory acceptable to the Government indicating compliance with the applicable requirements of UL Standard No. 62 and No. 471.

3.2.2 NSF International. Acceptable evidence of meeting the requirements of NSF International Standard No. 2 shall be a listing in the current edition of the NSF International "Listing of Food Service Equipment" and display of the NSF International mark on the finished product, or a certified test report from a recognized independent laboratory acceptable to the Government indicating compliance with the applicable NSF International requirement(s).

3.2.3 National Electrical Manufacturers Association (NEMA). The contractor shall submit to the contracting officer a written certificate of compliance as proof of compliance that the sandwich unit meets the applicable requirements of NEMA WD-1.

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4. PACKAGING

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each unit shall be cleaned, preserved, and packed in accordance with ASTM D 3951.

4.3 Export packaging and packing.

4.3.1 Export packaging. Each sandwich unit shall be cleaned, preserved, and cushioned to afford protection against corrosion, deterioration, and physical damage during shipping and storage.

4.3.2 Export packing. Each sandwich unit shall be packed in a snug fitting, wood cleated plywood box conforming to PPP-B-601. The sandwich unit shall be cushioned, blocked, and braced to provide a minimum of 1/2 inch clearance between the casters and the base of the box to prevent movement and damage during transit. Each box shall be closed and strapped in accordance with PPP-B-601.

5. NOTES

5.1 Intended use. The sandwich units covered by this commercial item description are intended for use on the serving line in dining halls.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.
b. PIN designating type, style and size required (see 1.1, 1.2, and 5.4).
c. Whether tray rail is required (see 2.3).
d. Electrical requirements if other than those specified (see 2.5).
e. Government approved manual number to be included on data nameplate (see 2.5).
f. Selection of applicable packaging and packing required (see 4.1).

5.3 Sources for documents.

5.3.1 Sources for non-Government association documents.

NEMA Standard No. WD-1 - General Requirements for Wiring Devices

NEMA documents are available from:

National Electrical Manufacturers Association (NEMA)
2101 L Street, NW
Washington, DC 20037

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NSF International Standard No. 2 - Food Service Equipment

NSF International documents are available from:

NSF International
3475 Plymouth Road
P.O. Box 130140
Ann Arbor, MI 48113-0140

UL Standard No. 62 - UL Standard for Safety Flexible Cord and Fixture Wire
UL Standard No. 471 - UL Standard for Safety Commercial Refrigerators and Freezers

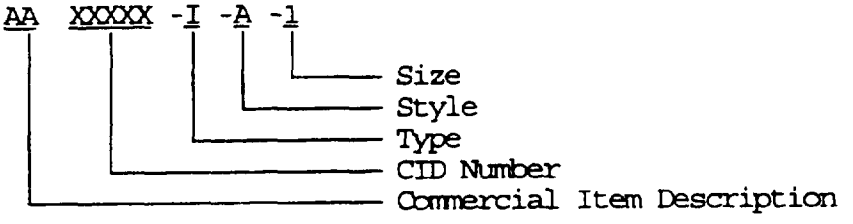
UL Standards are available from:

Underwriters Laboratories, Inc. (UL)
333 Pfingsten Road
Northbrook, IL 60062

5.3.2 Source for Government documents. Copies of military and federal documents are available from:

Standardization Documents Order Desk
Building 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

5.4 Part identification number (PIN). The PIN's to be used for items required by this CID are created as follows:



5.5 "Government unique" requirements. Whenever a "Government unique" requirement is included in the title of a paragraph under "Salient Characteristics", it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

5.6 Ozone depleting substances. For current information about the status of the Environmental Protection Agency's (EPA) Significant New Alternatives Policy (SNAP) program and specific refrigerant alternatives, call the Stratospheric Ozone Protection Hotline at (800) 296-1996 or the SNAP Coordinator at (202) 233-9195. For published information see the EPA

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Stratospheric Ozone Protection, Fact Sheet (EPA-430-F-93-004 dated February 1993) Short List of Alternative Refrigerants and Action Guide (EPA-430-F-93-003, dated February 1993) Cooling & Refrigerating Without CFCs.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 99

Review Activities

Army - MD, QM
Navy - CG
Air Force - 50, 84
DLA - GS

User Activity

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project 7310-0825

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial 508-651-5175	(2) AUTOVON/DSN 256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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